

DNS Analysis & APT Attack Chains

DNS Footprinting

What is DNS Footprinting? Information gathering process using DNS records during the reconnaissance phase. There is no direct hacking involved; the goal is to collect maximum infrastructure information silently.

What Attackers Try to Find:

- **Subdomains:** e.g., admin.company.com, dev.company.com, vpn.company.com
- Mail servers (MX Records)
- Public IP addresses
- Internal services & Network infrastructure layout

Impact:

- Complete infrastructure leakage
- Exposure of internal naming conventions
- Easier attack planning for adversaries

SOC Detection Indicators:

- High DNS query volume from one IP address
- Sequential subdomain queries / Brute-force DNS enumeration
- External IP querying many domains quickly

Real SOC Example:

- **Source IP:** 45.12.x.x
- **Queries:** admin.company.com, vpn.company.com, mail.company.com, dev.company.com
- **Conclusion:** Active DNS reconnaissance detected.

DNS Poisoning

What is DNS Poisoning? Also called DNS Cache Poisoning. It involves fake DNS records being inserted into the DNS cache, redirects users to malicious websites.

Main Goal: Credential theft, phishing, malware delivery, and malicious traffic redirection.

Normal DNS Flow:

User Request → Resolver → Root Server → TLD Server → Authoritative DNS → Correct IP Returned → Legitimate Website

DNS Poisoning Attack Flow:

Attacker Sends Fake Responses → Resolver Accepts → Malicious IP Cached → Users Redirected to Attacker Site

Technical Working:

The attacker triggers a DNS query, causing the resolver to forward the request. The attacker then floods fake responses, guessing the Transaction ID (TXID) and source port. If the fake response is accepted, the poisoned cache is used for future requests.

Types of DNS Poisoning:

1. **Cache Poisoning:** Fake records inserted into a shared DNS cache (Most dangerous type).
2. **Hosts File Poisoning:** Local hosts file modified; domain redirected locally.
3. **Router DNS Poisoning:** Router DNS settings changed; all devices use the malicious DNS server.

Example Scenario: User visits a banking website → DNS resolves to the attacker's IP → Fake banking page is shown → Credentials stolen.

Impact: Credential theft, malware infection, financial loss, and reputation damage.

Prevention:

Enable DNSSEC | Use trusted DNS resolvers | Patch DNS servers regularly | Deploy IDS/Firewall | Monitor DNS traffic | Secure routers | User awareness training

SOC Indicators:

- Domain resolves to an unknown IP or sudden DNS IP changes
- SSL certificate warnings
- Multiple failed DNS queries or strange external DNS communications

Quick SOC Checklist:

1. Verify domain → IP mapping
2. Check authoritative DNS source
3. Analyze DNS query patterns
4. Cross-check IP reputation

Real SOC Scenario:

- **Users Report:** "Google looks different."
- **DNS Logs Show:** google.com resolving to an unknown IP.
- **Conclusion:** DNS poisoning attack.

DNS Footprinting vs DNS Poisoning

Feature	Footprinting	Poisoning
Stage	Reconnaissance	Active Attack
Goal	Gather Information	Redirect Traffic
Visibility	Low	Medium
Impact	Infrastructure Exposure	Credential / System Compromise

SOC Mindset — Always ask:

- Why is this domain being queried?
- Is the pattern normal?
- Is the returned IP correct?

APT (Advanced Persistent Threat)

What is an APT? A long-term targeted cyber attack where attackers stay hidden inside a network for months or years.

Meaning of APT:

- **Advanced:** Uses sophisticated tools, custom malware, zero-days, and advanced evasion techniques.
- **Persistent:** Maintains long-term access, surviving system reboots and remediation.
- **Threat:** Highly organized attackers, such as nation-state or elite cybercriminal groups.

Full APT Attack Chain

1. **Reconnaissance:** Gather information about the target. (e.g., DNS enumeration, WHOIS lookup, LinkedIn research)
2. **Weaponization:** Create a malicious payload. (e.g., RAT malware, malicious Office documents, custom exploits)
3. **Delivery:** Send the payload to the victim. (e.g., Phishing emails, malicious links, USB attacks)
4. **Initial Access:** Gain the first foothold. (e.g., Exploiting vulnerabilities, macro execution, stolen credentials)
5. **Execution:** Run malicious code. (e.g., PowerShell, CMD/Bash, LOLBins)
6. **Persistence:** Maintain long-term access. (e.g., Registry Run Keys, Scheduled Tasks, WMI subscriptions)
7. **Privilege Escalation:** Gain higher privileges. (e.g., Token impersonation, local exploits, admin credential abuse)
8. **Credential Access:** Steal credentials. (e.g., Mimikatz, LSASS dumping, keylogging)

9. **Discovery:** Explore the internal network. (e.g., AD enumeration, network scanning, share discovery)
10. **Lateral Movement:** Move to other systems. (e.g., RDP, SMB, PsExec, Pass-the-Hash)
11. **Command & Control (C2):** Communicate with the attacker server. (e.g., HTTPS beaconing, DNS tunneling, encrypted traffic)
12. **Exfiltration & Impact:** Steal or destroy data. (e.g., Data exfiltration, ransomware, wipers)

Common APT Indicators (SOC View):

- Unusual outbound connections • New/untrusted domains contacted • Abnormal login timings
- One user accessing many systems • Large outbound data transfer • Heavy PowerShell/WMI usage
- Security tools disabled • Credential dumping activity • New persistence mechanisms

APT Defense Best Practices:

Patch systems regularly | Use email filtering | Monitor endpoints and logs | Deploy EDR/XDR | Apply least privilege | Segment networks | Conduct awareness training | Maintain incident response plans

APT vs Normal Attack

Feature	Normal Attack	APT
Duration	Short-term	Long-term
Goal	Quick gain	Strategic espionage
Detection	Easier	Difficult
Behavior	Noisy	Stealthy

Final SOC Analyst Mindset

As a SOC Analyst: Never analyze alerts in isolation. Connect events together, look for attack chains, and check behavior patterns.

Focus on: Who? • Why? • How? • What happened next?